

STANES SCHOOL

1044, Avinashi Road, Coimbatore – 18.

UNIT TEST – 1 2026-2027

Class : XII

Subject : Informatics Practices

Duration : 2 Hours

Max. Marks : 50

SECTION A (17 × 1 = 17 Marks)

1. State whether the following statement is **True** or **False**:

A relation can have multiple primary keys.

2. Which of the following represents a **tuple** in a relational database?

- a) A column in a table
- b) A row in a table
- c) A database name
- d) A data type

3. What is the full form of **DDL** in SQL?

- a) Data Definition Language
- b) Data Display Language
- c) Database Definition Language
- d) Domain Definition Language

4. Which SQL command is used to **remove an entire table** from the database?

- a) DELETE
- b) DROP
- c) REMOVE
- d) TRUNCATE

5. Which of the following is **not** a valid SQL data type?

- a) INT
- b) VARCHAR
- c) DATE
- d) LIST

6. The **domain** of an attribute in a relational database refers to:

- a) The name of the column
- b) The set of allowed values for that column
- c) The primary key of the table
- d) The number of rows in the table

7. State whether the following statement is **True** or **False**:

SELECT * FROM Employees; → this command will display all columns from the Employees table.

8. Which SQL clause is used to **filter rows** based on a condition?

- a) GROUP BY
- b) ORDER BY
- c) WHERE
- d) HAVING

9. Which of the following is a **valid relational operator** used in SQL WHERE clause?

- a) ==
- b) <>
- c) :=
- d) != =

10. Which keyword is used to **eliminate duplicate rows** in the output of a SELECT query?

- a) UNIQUE
- b) DISTINCT
- c) DIFFERENT
- d) NO DUPLICATE

11. Which of the following commands is used to **add a new column** to an existing table?

- a) ADD COLUMN
- b) ALTER TABLE ... ADD
- c) CREATE COLUMN
- d) INSERT COLUMN

12. Which of the following is **not** a valid logical operator in SQL?

- a) AND
- b) OR
- c) NOT
- d) WITH

13. What does the **BETWEEN** operator do in SQL?

- a) Checks if a value is within a range
- b) Checks if a value is NULL
- c) Checks if two values are equal
- d) Checks if a value matches a pattern

14. **Assertion (A):** A candidate key can uniquely identify each row in a table.

Reason (R): Every table must have at least one candidate key.

Choose the correct option:

- a) Both A and R are True and R is the correct explanation for A
- b) Both A and R are True but R is not the correct explanation for A
- c) A is True but R is False
- d) A is False but R is True

15. Which SQL command is used to **create a new database**?

- a) CREATE TABLE
- b) CREATE DATABASE
- c) NEW DATABASE
- d) ADD DATABASE

16. State whether the following statement is **True** or **False**:

In a relational database, each row in a table is called an attribute.

17. Which of the following is a **valid SQL string data type**?

- a) INT
- b) DATE
- c) VARCHAR
- d) FLOAT

SECTION B (5 × 2 = 10 Marks)

18. **A.** Define **Database Management System (DBMS)**. Mention any two advantages of using DBMS over traditional file systems.

OR

B. Differentiate between **Data Definition Language (DDL)** and **Data Manipulation Language (DML)** with one example of each.

19. Consider the following table **Employee**:

EmpID	EmpName	Department	Salary
101	Aarav	IT	75000
102	Priya	HR	68000
103	Rohan	IT	85000

- Write SQL queries for the following:
- I. Display the names of all employees working in the **IT** department.
 - II. Display the **EmpID** and **EmpName** of employees whose salary is greater than 70000.
- 20. A.** What is a **Candidate Key**? Explain with an example.

OR

B. Differentiate between **Primary Key** and **Alternate Key**.

- 21.** Write SQL statements for the following tasks:
- I. Create a database named **SchoolDB**.
 - II. Create a table **Student** with the following structure:

Column Name	Data Type
RollNo	INT
Name	VARCHAR(30)
Class	INT
Marks	FLOAT

22. Consider the table **Products**:

ProdID	ProdName	Price
1	Laptop	55000
2	Mouse	500
3	Keyboard	1200
4	Monitor	8000

- Write the output of the following SQL queries:
- I. SELECT * FROM Products WHERE Price BETWEEN 1000 AND 10000;
 - II. SELECT ProdName FROM Products WHERE Price > 5000 AND Price < 50000;

SECTION C (2 × 3 = 6 Marks)

- 23. A.** Explain the following **Relational Data Model** terms with suitable examples:
- I. Domain
 - II. Tuple
 - III. Relation

OR

B. What are **Alternate Keys**? How are they different from **Candidate Keys**? Explain with an example.

24. A. Consider the table **Library**:

BookID	Title	Author	Price
101	Python Basics	Rajesh	450
102	SQL Guide	Priya	350
103	Data Science	Amit	600
104	Web Dev	Neha	500
105	AI Fund	Rajesh	550

- Write SQL queries for the following:
- I. Display all books whose price is greater than 400 and less than or equal to 550.
 - II. Display the titles of books written by **Rajesh**.
 - III. Display all books sorted by **Price** in descending order.

OR

- B.** Write the output of the following SQL queries based on the **Library** table above:
- I. SELECT * FROM Library WHERE Author = "Rajesh" AND Price > 500;
 - II. SELECT DISTINCT Author FROM Library;
 - III. SELECT Title, Price FROM Library WHERE Price BETWEEN 400 AND 550;

SECTION D (3 × 4 = 12 Marks)

25. Consider the following table **Sales**:

SaleID	ProductName	Category	Quantity	PricePerUnit
101	Laptop	Electronics	5	50000
102	Mouse	Electronics	10	500
103	Pen	Stationery	50	20
104	Keyboard	Electronics	3	1200
105	Notebook	Stationery	30	40

- Write SQL queries for the following:
- I. Display all records where **Category** is "Electronics".
 - II. Display **ProductName** and **Quantity** for products with Quantity greater than 5.
 - III. Display all products where **PricePerUnit** is less than 1000.
 - IV. Display all products sorted by **ProductName** in alphabetical order.

26. A. Write SQL statements to perform the following operations:

Column Name	Data Type
EmpID	INT
EmpName	VARCHAR(35)
Department	VARCHAR(20)
Salary	DECIMAL(10,2)

- I. Create a database named **CompanyDB**.
- II. Create a table **Employee** with the following structure:
- III. Add a new column **City** of type VARCHAR(25) to the Employee table.
- IV. Change the data type of **Salary** column to INT.

OR

B. Write SQL queries for the following based on the **Employee** table created above (assuming the table has some data):

- I. Display all employees whose **Department** is either "IT" or "HR".
- II. Display the **EmpName** and **Salary** of employees with Salary between 40000 and 70000.
- III. Display all columns for employees whose name starts with the letter 'A' .
- IV. Display all employees sorted by **Salary** in descending order.

27. Predict the output of the following SQL queries based on the **Student** table shown below:

Table: Student

RollNo	Name	City	Marks
1	Aarav	Delhi	85
2	Priya	Mumbai	92
3	Rohan	Delhi	78
4	Simran	Pune	88
5	Karan	Mumbai	75

Write the output for each query:

- I. SELECT DISTINCT City FROM Student;
- II. SELECT Name, Marks FROM Student WHERE Marks > 80 AND City = "Delhi";
- III. SELECT * FROM Student WHERE City IN ("Mumbai", "Pune");
- IV. SELECT * FROM Student WHERE City = "Delhi" OR Marks > 90;

SECTION E (1 × 5 = 5 Marks)

28. A. Write suitable SQL queries for the following tasks:

- I. To display the first three characters of the **ProductName** column from the **Products** table.
- II. To display the **Name** column from the **Customers** table after converting it to uppercase.
- III. To display the **Address** column from the **Customers** table.
- IV. To display the **current date**.
- V. To display all columns from the **Orders** table where the **OrderDate** is after '2025-01-01'.

OR

B. Consider the table **Employees** with columns: **EmpID, EmpName, Department, Salary, JoinDate**.

Write SQL queries for the following:

- I. Display all employees whose **Department** is "IT" and **Salary** is greater than 60000.
- II. Display the **EmpName** and **JoinDate** of employees who joined after **2023-01-01**.
- III. Display all employees whose **Department** is either "IT" or "Finance".
- IV. Display all employees sorted by **EmpName** in ascending order.
- V. Display all employees whose **EmpName** starts with the letter 'A'.